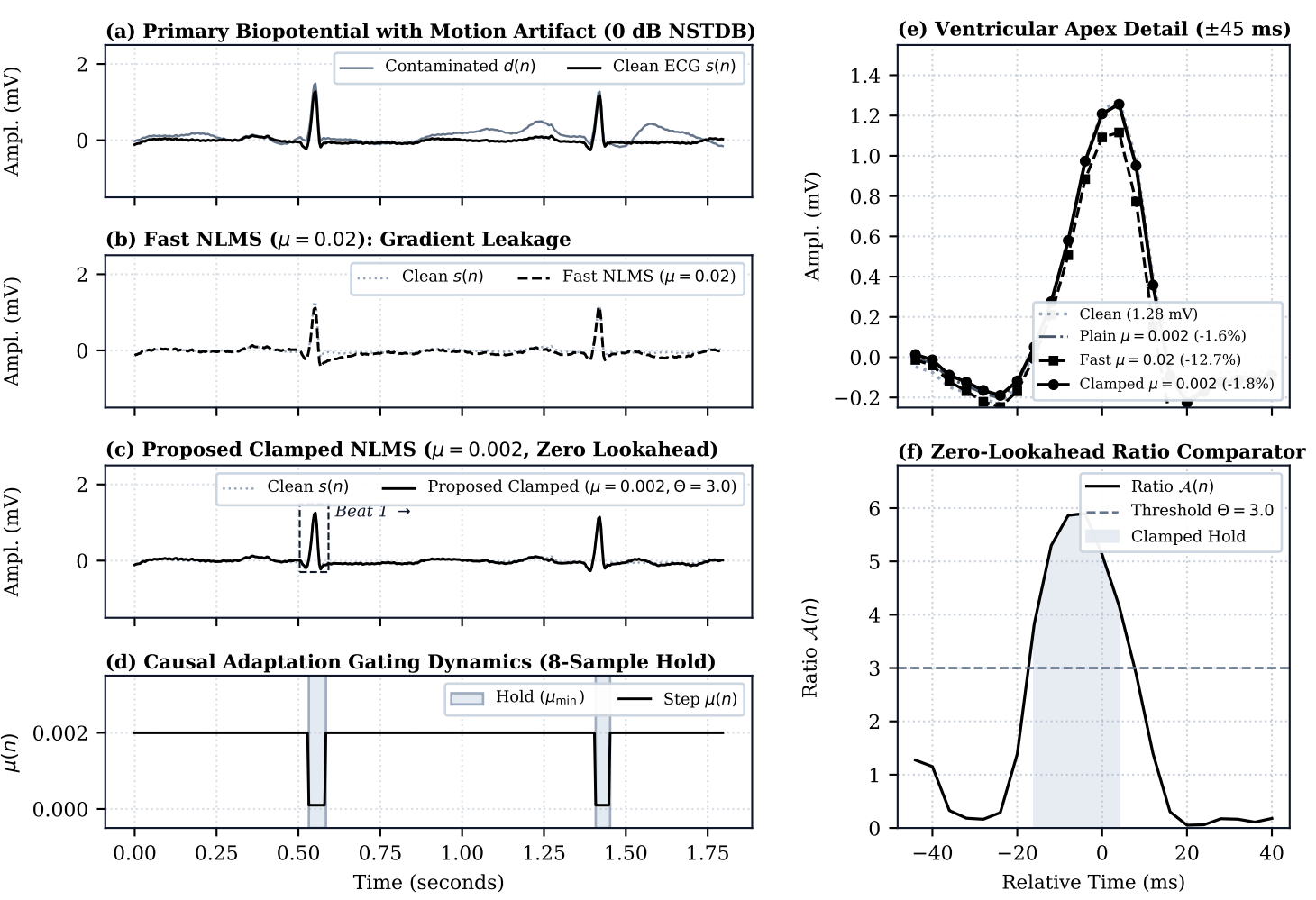

(a) Primary Biopotential with Motion Artifact (0 dB NSTDB)
Contaminated $d(n)$
Clean ECG $s(n)$
Ampl. (mV)
(b) Fast NLMS ($\mu = 0.02$): Gradient Leakage
Clean $s(n)$
Fast NLMS ($\mu = 0.02$)
Ampl. (mV)
(c) Proposed Clamped NLMS ($\mu = 0.002$, Zero Lookahead)
Clean $s(n)$
Proposed Clamped ($\mu = 0.002, \Theta = 3.0$)
Beat 1 →
Ampl. (mV)
(d) Causal Adaptation Gating Dynamics (8-Sample Hold)
Hold ($\mu_{\min}$)
Step $\mu(n)$
$\mu(n)$
0.002
0.000
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Time (seconds)
(e) Ventricular Apex Detail (±45 ms)
Ampl. (mV)
Clean (1.28 mV)
Plain $\mu = 0.002$ (-1.6%)
Fast $\mu = 0.02$ (-12.7%)
Clamped $\mu = 0.002$ (-1.8%)
(f) Zero-Lookahead Ratio Comparator
Ratio $\mathcal{A}(n)$
Threshold $\Theta = 3.0$
Clamped Hold
Ratio $\mathcal{A}(n)$
−40
−20
0
20
40
Relative Time (ms)